# Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

by


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DEPARTMENT OF CIVIL ENGINEERING


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# Candidate's Declaration

This is to certify that the work presented in this thesis entitled, **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"**, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

**Signature of the Candidates**

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# Recommendation Letter from Thesis Supervisor

The thesis entitled **“Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)”** submitted by the students:

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2. Md Sagor Hossain Saddas (Reg. No.: 2020333553)

is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor’s Degree.

______________________________

Signature of the Supervisor
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**Date:** ______________________________

# Certificate of Acceptance

The thesis titled **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"** submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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**The Authors**

## Abstract

**Keywords:**

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# 1 Introduction

The performance of flexible pavements is fundamentally governed by the quality of the bituminous binder and the physical characteristics of the constituent aggregates [2]. In the context of Bangladesh's expanding road network, flexible pavements constitute the majority of the infrastructure, supporting rapid urbanization and increased traffic volumes. Bitumen acts as the primary binding agent, providing structural integrity and flexibility to withstand dynamic traffic loads and environmental stressors. However, the subtropical climate of Bangladesh, characterized by high temperatures and heavy monsoon rainfall, imposes severe demands on pavement materials, often leading to premature distress such as rutting and stripping [**?**].

To meet the growing demand for road construction, Bangladesh relies on a combination of locally produced and imported bitumen. The state-owned Eastern Refinery Limited (ERL) produces laboratory-grade bitumen, which is widely regarded for its consistency and adherence to standard specifications. Conversely, a significant portion of the market is supplied by imported low-cost bitumen to offset production shortfalls. The variation in the penetration grade and temperature susceptibility of these different binder sources plays a critical role in the long-term performance and climate resilience of the constructed pavements [**?**].

Despite the widespread use of imported bitumen, quality inconsistencies have raised significant concerns regarding its long-term viability. Empirical evidence suggests that adulterated or substandard imported bitumen contributes to rapid pavement deterioration and early rutting [**?**]. Consequently, agencies such as the Roads and Highways Department (RHD) enforce strict testing protocols and have occasionally restricted certain low-cost imported variants. In contrast, locally produced ERL bitumen has demonstrated superior performance in comparative evaluations and maintains a stable pricing structure, although systematic economic and mechanical trade-off analyses remain limited [**?**]. A critical gap remains in understanding the comparative life-cycle performance of these two primary binder sources under localized traffic and climatic conditions.

Furthermore, the mechanical response of bituminous mixtures is significantly dictated by the shape and gradation of the coarse aggregates. Prior investigations have demonstrated that replacing flaky and elongated aggregates with cubical aggregates enhances the interlocking mechanism, thereby increasing the Marshall stability and rutting resistance of the mix [3, 4]. While the influence of aggregate shape factors is well-documented [1], no prior study has systematically compared the synergistic effects of substituting conventional aggregates with specifically proportioned cubical aggregates across different bitumen sources in Bangladesh.

This study aims to systematically evaluate and compare the fundamental properties, mix design performance, and cost-effectiveness of local laboratory-grade bitumen and imported low-cost bitumen, focusing on the influence of coarse aggregate shape. The specific objectives include assessing standard binder properties, investigating the effect of substituting 20% of the coarse aggregate fraction (19 mm to 12.5 mm) with cubical aggregates on Marshall stability and flow, evaluating climate resistance via modified Marshall immersion, and performing a comparative economic analysis. The findings are expected to inform RHD specifications and promote sustainable road construction practices in Bangladesh. Section 2 reviews the existing literature; Section 3 details the experimental methodology; Section 4 presents the results and discussion; and Section 5 concludes the study with practical recommendations.

# 2 Literature Review

## 2.1 Bitumen Grading and Properties in Tropical Climates

The structural durability of flexible pavements in tropical climates is heavily dependent on the rheological properties of the bituminous binder under high ambient temperatures [?, ?]. In regions like Bangladesh and India, where pavement surface temperatures can reach up to 60°C during the summer, the binder softens significantly, rendering the pavement susceptible to permanent deformation such as rutting and shoving [?, ?]. Historically, the penetration grading system (e.g., 60/70 or 80/100 grades) was the standard specification for bitumen selection in these areas [?, ?]. However, several investigations have demonstrated that this empirical method fails to directly measure fundamental temperature susceptibility, making it inadequate for predicting pavement performance under extreme thermal variations [?, ?]. Consequently, recent research advocates for the adoption of performance-based specifications, such as Viscosity Grading (VG) and Superpave Performance Grading (PG), to ensure the binder maintains sufficient stiffness during peak high temperatures [?, ?].

## 2.2 Performance Comparison of Local vs. Imported Bitumen

To meet the escalating demand for infrastructure development, Bangladesh relies on a combination of locally produced laboratory-grade bitumen from Eastern Refinery Limited (ERL) and various imported low-cost variants [?, ?]. Empirical evidence suggests that while local ERL bitumen adheres strictly to standardized penetration and viscosity specifications, the quality of imported bitumen is often inconsistent due to a lack of rigorous, localized quality certification upon arrival [?, ?]. Furthermore, these inconsistencies have been directly linked to premature pavement failures, as substandard imported binders lack the requisite elastic recovery and rutting resistance for tropical monsoon conditions [?, ?]. While some studies suggest that imported bitumen can perform adequately if strictly tested and modified with polymers [?, ?], a comprehensive mechanical and rheological comparison between local ERL bitumen and specific low-cost imported alternatives remains scarce in the current literature.

## 2.3 Aggregate Shape and its Influence on Mix Design

The mechanical response and load-carrying capacity of bituminous mixtures are fundamentally governed not only by the binder but also by the geometric characteristics of the coarse aggregates [?, 4]. Prior investigations have established that aggregate shape factors—specifically flakiness and elongation indices—play a critical role in determining the structural skeleton of the mix [?, 1]. Flaky and elongated particles are structurally deficient; they tend to fracture easily during the compaction process and fail to provide adequate particle-to-particle interlocking [?, 3]. Conversely, cubical and angular aggregates exhibit higher internal friction, creating a robust interlocking matrix that effectively resists plastic flow under dynamic traffic loads [?, ?]. The deliberate substitution of conventional aggregates with precisely proportioned cubical aggregates has been shown to significantly enhance the shear strength of the resulting pavement layer [?, 4].

## 2.4 Marshall Mix Design and Performance Evaluation

The Marshall mix design method remains the predominant framework for evaluating the volumetric and mechanical properties of asphalt mixtures in developing countries [?, 2]. Researchers extensively utilize parameters such as Marshall stability, flow value, and voids in mineral aggregate (VMA) to quantify how constituent materials affect overall strength [?, ?]. Mixes incorporating higher proportions of cubical aggregates consistently demonstrate superior Marshall stability values and optimal volumetric structures, which directly translate to improved long-term durability [?, 3]. Furthermore, in regions experiencing heavy monsoon rainfall, the moisture susceptibility of the mix—often evaluated through modified Marshall immersion tests—is critically influenced by both the aggregate surface texture and the binder's adhesion properties [?, ?]. Although individual components have been thoroughly evaluated, the synergistic effect of integrating cubical aggregates with different grades and sources of bitumen requires further systematic exploration [?, ?].

## 2.5 Cost-Effectiveness Studies in Pavement Engineering

Economic constraints in developing nations mandate a rigorous analysis of pavement construction materials, heavily weighing initial capital expenditures against long-term life cycle costs (LCC) [?, ?]. Flexible pavements are overwhelmingly favored in Bangladesh due to their lower initial construction costs compared to rigid concrete pavements [?, ?]. However, life cycle cost analyses indicate that frequent periodic maintenance—necessitated by premature distress from overloaded vehicles and climatic factors—often negates these initial savings [?, ?]. To improve the cost-effectiveness of flexible pavements, engineers are increasingly turning toward strategic material selection, such as utilizing higher-quality local bitumen or introducing specific aggregate shapes to extend the pavement's service life without incurring the substantial upfront costs of rigid pavements [?, ?]. Ultimately, optimizing the binder source and aggregate geometry represents a highly sustainable and economically viable approach to infrastructure development [?, ?].

## 2.6 Critical Gap Statement

Despite extensive research on the independent effects of bitumen grading in tropical climates [?, ?] and the influence of aggregate shape on mix performance [?, 4], a critical gap remains in the literature. No prior study has systematically compared the mechanical performance and cost-effectiveness of local laboratory-grade bitumen against imported low-cost bitumen while simultaneously evaluating the enhancing effects of a 20% cubical aggregate substitution. This study seeks to bridge this gap, providing essential empirical data to optimize flexible pavement design in Bangladesh.

# 3 Methodology

# 4 Results and Discussion

# 5 Conclusion

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